

Program Structure and Syllabus of B. Tech II Year

Information Technology

(Academic Regulations - AR24)



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B. TECH II YEAR I SEMESTER

S.No	Course Code	Category	Course	Hours per week			Credits
				L	T	P	
1	EMA2101	Theoretical (PC)	Computer Organization	3	0	0	3
2	EMA2X02	Theoretical (ES)	Discrete Mathematics	3	0	0	3
3	EMD2X01	Theoretical (BS)	Computer Oriented Statistical Methods	3	0	0	3
4	EMA2103	Theoretical & Practical (PC)	Data Structures and Algorithms	3	0	4	5
5	EMA2104	Theoretical & Practical (PC)	Object Oriented Programming through Java	3	0	4	5
6	EVA2101	Exploratory (PC)	Integrated Project - I	0	0	2	1
TOTAL				15	0	10	20

B. TECH II YEAR II SEMESTER

S.No	Course Code	Category	Course	Hours per week			Credits
				L	T	P	
1	EMA2201	Theoretical (PC)	Operating Systems	3	0	0	3
2	EMA2202	Theoretical (PC)	Design and Analysis of Algorithms	3	0	0	3
3	EMA2X03	Theoretical (PC)	Software Engineering	3	0	0	3
4	EMA2204	Theoretical & Practical (PC)	Database Systems	3	0	2	4
5	EMA2205	Theoretical & Practical (PC)	Advanced Java Programming	2	0	4	4
6	EAE2221	Practical (HS)	English through Theatre Arts	1	0	2	2
7	EVA2201	Exploratory (PC)	Integrated Project - II	0	0	2	1
8	--	IN*	Summer Internship / Certification	0	0	0	0
TOTAL				14	0	10	20

*Summer Internship / Certification will begin during the semester break just after the completion of the 4th semester and evaluation shall be done during 5th semester.

Suggested activities: Hands-on Training/Short Research Project, Seminar attendance, Certification from Swayam or NPTEL or Coursera (which shall follow a proctored assessment) read assigned journals to prepare for seminars, Study certain entrepreneurs, social projects, etc.

B. TECH III YEAR I SEMESTER

S.No	Course Code	Category	Course	Hours per week			Credits
				L	T	P	
1		PC	Data Communications and Networking	3	0	4	5
2		PC	Python Programming	3	0	4	5
3		PE-I	1. Object Oriented Analysis and Design	3	0	0	3
			2. Formal Languages and Automata Theory				
			3. Software Testing				
4		OE-I	1. Foreign Language (Japanese/ German/ French)	2	0	0	2
			2. Content Writing				
			3. Personality Development and Life Skills				
5		PC	Web Technologies	0	0	4	2
6		BS	Quantitative Aptitude and Reasoning	0	0	2	1
7		IN	Summer Internship	0	0	4	2
TOTAL				11	0	18	20

B. TECH III YEAR II SEMESTER

S.No	Course Code	Category	Course	Hours per week			Credits
				L	T	P	
1		PC	Cloud Application Architectures	3	0	4	5
2		PC	Data Science	3	1	0	4
3		PE-II	1. Mobile Application Development	3	0	4	5
			2. Machine Learning and its Applications				
			3. Cryptography and Network Security				
4		PE-III	1. Fundamentals of Cyber Security	3	1	0	4
			2. Compiler Design				
			3. Wireless Network Security				
5		BS	Verbal Ability	0	0	2	1
6		BS	Critical Reasoning	0	0	2	1
7		IN	Industry Oriented Mini Project	0	0	0	0
TOTAL				12	2	12	20

B. TECH IV YEAR I SEMESTER

S.No	Course Code	Category	Course	Hours per week			Credits
				L	T	P	
1		PC	Predictive Analytics using R Programming	3	0	4	5
2		PC	Artificial Intelligence	3	1	0	4
3		PE-IV	1. Blockchain Technology	3	0	4	5
			2. Big Data Analytics				
			3. Secure Software Development				
4		HS	Financial Literacy	2	0	0	2
5		PC	Comprehensive Viva-Voce	0	0	4	2
6		IN	Industry Oriented Mini Project	0	0	4	2
TOTAL				11	1	16	20

B. TECH IV YEAR II SEMESTER

S.No	Course Code	Category	Course	Hours per week			Credits
				L	T	P	
1		OE-II	1. Narrative Analytics	3	0	0	3
			2. Management Science				
			3. Workplace Communication for Engineers				
			OR 4. Industry Elective (MOOCs)+				
2		OE-III	1. Sustainable Development Goals (MOOCs)	3	0	0	3
			2. Intellectual Property Rights				
			3. Approaches to critical thinking				
			OR				
			4. Industry Elective (MOOCs)+				
3		PR	Technical Seminar	0	0	4	2
4		PR	Project	0	0	24	12
TOTAL				6	0	28	20

+Students who are shortlisted for industry electives, it is mandatory for them to register for the industry elective course in lieu of the one of the two open elective courses (OEC-II and OEC-III), i.e., such students shall have to register either OEC-II or OEC-III in addition to the industry elective course. For those who are not shortlisted for the industry elective are required to register OEC-II and OEC-III.

Computer Organization

B. Tech II Year I Semester					Dept. of Information Technology			
Code	Category	Hours / Week			Credits	Marks %		
EMA2101	Theoretical (PC)	L	T	P	C	CIE	SEE	Total
		3	0	0	3	50	50	100

Course Objectives

Course Objectives of Computer Organization are to:

1. Demonstrate different types of Instructions and addressing modes
2. Compare different Modes of Transfer
3. Summarize the concepts of memory organization
4. Outline Performance of cache memory.
5. Illustrate the methods used in microprogram sequencing

Course Outcomes

At the end of this Computer Organization course, students will be able to:

1. Analyze Instructions and Instruction Sequencing
2. Examine how interrupts contribute to optimized and efficient I/O operations.
3. **Compare** memory mapping techniques
4. **Analyze** the role of secondary storage in the memory hierarchy
5. Investigate how a hardwired control unit operates.

UNIT I

Basic Structure of Computers: Computer types, Functional Units, Basic Operational Concepts, Bus Structures, and Software. [Ch1:1.1- 1.5, Book1]

Machine Instructions: Instructions and Instruction Sequencing: Register transfer notation, Assembly language notation, Basic instruction types, Instruction Execution and straight-line sequencing. Branching, Condition Codes. Generating Memory Address.

Addressing Modes: Implementation of Variables and constants, Indirection and pointers, Indexing and Arrays, Relative Addressing, Additional Modes. [Ch 2:2.4, 2.5, Book1]

UNIT II

Input – Output Organization: Accessing I/O Devices Interrupts with examples, Interrupt Hardware, Enabling and disabling interrupts, Handling Multiple Devices, Controlling Device requests, Exceptions. Direct memory Access, Bus arbitration, Buses, Synchronous Bus, Asynchronous bus. [Ch4:4.1, 4.2, 4.4, 4.5, Book1]

UNIT III

Memory System: Basic Concepts, Semiconductor RAM memories, internal organization of memory chips. Static memory, Asynchronous DRAM, Synchronous DRAM, Structure of larger Memories. Memory System Consideration, Rambus Memory. Read Only Memories and Memory Hierarchy. Cache Memory, Mapping Function (Direct, Associative and Set Associate Mapping). Replacement Algorithms. Example of mapping Techniques. [Ch5: 5.1-5.5, Book1]

UNIT IV

Performance Consideration, Interleaving, Hit Ratio and Miss Penalty, Cache on the Processor Chip, Other enhancements. Virtual Memories, Address Translation, Memory Management Requirements, Secondary Storage, Magnetic Hard Disks, Optical Disks, Magnetic Tape Systems. [Ch5: 5.6-5.9, Book1]

UNIT V

Basic Processing Unit: Register Transfer, Performing an Arithmetic or Logic operations, fetching a

word from memory, storing a word in Memory. Execution of a complete Instruction, Branch Instructions. Multiple Bus Organization. Hardwired Control, A complete Processor, Microprogrammed Control, Microinstructions, Microprogram Sequencing, Wide-Branch addressing, Microinstruction with Next-Address Field, Prefetching microinstructions, Emulation. [Ch7, Book1]

Text Books:

1. Carl Hamacher, Zvonks Vranesic, SafeaZaky, Computer Organization, Fifth Edition, McGraw Hill, 2002.

Reference Books:

1. M. Morris Mano, Computer System Architecture, Third Edition, Pearson/PHI, 2011.
2. William Stallings, Computer Organization and Architecture, Sixth Edition, Pearson/PHI, 2007.

Discrete Mathematics

B. Tech II Year I Semester					Dept. of Information Technology			
Code	Category	Hours / Week			Credits	Marks %		
EMA2X02	Theoretical (ES)	L	T	P	C	CIE	SEE	Total
		3	0	0	3	50	50	100

Course Outcomes

Course Outcomes of Discrete Mathematics are to:

1. Interpret the Sets, syntax and semantics of propositional and predicate logic.
2. Solve applications involving Permutations and Combinations.
3. Formulate Recurrence relations to solve problems involving an unknown sequence.
4. Explain the concepts of Relations and Lattices.
5. Illustrate the Algebraic Systems.

Course Outcomes

At the end of this Discrete Mathematics course, students will be able to:

1. Distinguish between Statement Logic and Predicate Logic.
2. Apply the principles of Permutations and Combinations.
3. Solve Recurrence Relations by using generating functions.
4. Apply the knowledge of Relations and Lattice theory.
5. Analyze the Algebraic Systems with their properties.

UNIT I

Foundations: Basics, Sets and Operations of Sets, Fundamentals of Logic, Logical Inferences, First order logic and other methods of Proof, Rules of Inference for Quantified Propositions. [Ch1:1.1,1.2,1.5 -1.8, Book1]

UNIT II

Elementary Combinatorics: Basics of Counting, Combinations and Permutations, Enumerating Combinations and Permutations with & without repetitions, constrained repetitions, and Principle of Inclusion and Exclusion. [Ch2:2.1 - 2.5 & 2.8, Book1]

UNIT III

Recurrence Relations: Generating Functions, calculating coefficient of Generating Function, Solving Recurrence relations by substitution method and Generating Functions, The Method of Characteristic Roots, Solutions to inhomogeneous recurrence relations. [Ch3, Book1]

UNIT IV

Relations and Lattices: Relations, adjacency matrices and Directed Graphs, Operations on Relations, Special Properties of Binary Relations, Equivalence Relations, Ordering Relations, Lattices. [Ch4:4.1 to 4.5 & 4.7, Book1]

UNIT V

Algebraic structures: Algebraic systems, examples and general properties, semi groups and monoids, groups, sub groups, homomorphism, isomorphism, Permutation Groups and cyclic permutations. [Ch3:3.1,3.2,3.5.1 & 3.5.2 – Book2, Ch4: 4.7.5 -RBook2]

Text Books:

1. Joe L. Mott, Abraham Kandel, Theodore P. Baker, "Discrete Mathematics for Computer Scientists

and Mathematicians", Second Edition, PHI, 2019.

2. J. P. Tremblay and P. Manohar, "Discrete Mathematical Structures with Applications to Computer Science", Tata McGraw Hill, 2008.

Reference Books:

1. K. H. Rosen, "Discrete Mathematics and its Applications with Combinatorics and Graph Theory", 7th Edition, Tata McGraw Hill.
2. S. K. Chakraborty and B.K. Sarkar," Discrete Mathematics ", Oxford, 2011.
3. C. L. Liu and D. P. Mohapatra, "Elements of Discrete Mathematics-A Computer Oriented Approach", 3rd Edition, Tata McGraw Hill.

Computer Oriented Statistical Methods

B. Tech II Year I Semester					Dept. of Information Technology			
Code	Category	Hours / Week			Credits	Marks %		
EMD2X01	Theoretical (BS)	L	T	P	C	CIE	SEE	Total
		3	0	0	3	50	50	100

Course Objectives

Course Objectives of Computer Oriented Statistical Methods are to:

1. Understand various types of moments, Random Variables and Mathematical Expectation.
2. Understand the properties and applications of probability distributions.
3. Understand the Concept of Correlation, Regression and its measures, Understand Hypothesis Testing Framework.
4. Perform various t-tests, chi-square tests, F-tests and ANOVA.
5. Understand the concepts of Queuing Theory.

Course Outcomes

At the end of this Computer Oriented Statistical Methods course, students will be able to:

1. Define and differentiate between moments, random variables. Compute and interpret the mathematical expectation.
2. Calculate key measures such as **mean, variance** for Binomial, Poisson, Normal and Exponential distributions and their applications.
3. Compute, interpret the **Pearson, Spearman measures of correlation**. Fit and interpret a **simple linear regression model**. and applications of tests for simple mean and difference of means in large samples.
4. Formulate statistical hypotheses and apply appropriate hypothesis testing methods (t-test, chi-square test, ANOVA) to make data-driven decisions.
5. Applications of Queuing Theory in arrival and departure process.

UNIT I

Moments-Definition, Central and Non-Central, Skewness, Kurtosis (Based on Moments only). **Random Variables**: Definition of random variable, discrete and continuous random variables, probability mass function and probability density function with applications.

Mathematical Expectation: Definition of expectation, Variance, covariance and their properties with applications. Definition and properties of moment generating function.

[2.1.5-2.1.7, 5.2.1, 5.3.1, 5.3.2, 5.4.1-5. 4.3, 6.2-6.6, 7.1, 7.1.2, Book1]

UNIT II

Probability Distributions: Definitions, Derivation of mean and variance of Binomial, Poisson, Normal and Exponential Distributions and their applications.

[8.4.1, 8.4.6, 8.5.1, 8.5.5, 9.2.2, 9.2.5, 9.2.11, 9.8.1, Book1]

UNIT III

Correlation and Regression: Bivariate data, Concept of correlation, computation of Karl-Pearson correlation coefficient. Spearman's rank correlation coefficient. Simple linear regression, correlation verses regression, lines of regression and properties of regression coefficients. **Testing of Hypothesis**: Null and Alternative hypothesis, Critical region, two types of errors, Level of significance. One and two tailed tests. Procedure for testing of hypothesis. **Large Samples**: Tests for single sample mean, Difference of means,

single sample proportion, Difference of proportions.

[10.1, 10.2, 10.4, 10.7.1, 11.2.1-11.2.3, 14.4.1, 14.4.3-14.4.5, 14.7, 14.8.3, 14.8.4, Book1]

UNIT IV

Small Samples: Degrees of freedom, Tests of significance based on student's t-test for single sample specified mean, difference of means for independent and Paired t-test. Chi-Square test for Goodness of fit and Independence of attributes. F - test for equality of population variances. ANOVA-One way.

[15.6.2, 15.6.3, 16.2, 16.3.1-16.3.3, 16.6.1, Book-1, 5.1, 5.2, Book3]

UNIT V

Queuing Theory: Introduction-Queuing system-The arrival pattern-The service pattern-The queue discipline, Symbolic Representation of a Queuing Model –Characteristics of Infinite Capacity, Single server Poisson Queue Model, Queuing Problem-Pure Birth and Death Process-Probability Distribution of Departures (pure death process)- Basic Queuing Models-Measures of the $(M/M/1) : (\infty/FIFO)$ model-Characteristic of Finite Capacity.

[13.1-13.4, Book2]

Text Books:

1. V. K. Kapoor and S. C. Gupta: Fundamentals of Mathematical Statistics, Sultan Chand & Sons, New Delhi, eleventh edition.
2. Dr T.K.V. Iyengar & Dr B. Krishna Gandhi & S. Ranganadham & Dr M.V.S.S.N. Prasad: Probability and Statistics, Sultan Chand & Sons, New Delhi, seventh edition.
3. S. C. Gupta and V. K. Kapoor: Applied Statistics, Sultan Chand & Sons, New Delhi.

Reference Books:

1. Sheldon M Ross, Probability and statistics for Engineers and scientists, Academic Press.
2. T.T. Soong, Fundamentals of Probability and Statistics for Engineers, John Wiley & Sons Ltd, 2004.

Data Structures and Algorithms

B. Tech II Year I Semester					Dept. of Information Technology			
Code	Category	Hours / Week			Credits	Marks %		
EMA2103	Theoretical & Practical (PC)	L	T	P	C	CIE	SEE	Total
		3	0	4	5	50	50	100

Course Objectives

Course Objectives of Data Structures and Algorithms are to:

1. Appraise the fundamental concepts of trees.
2. Describe the various principles of BST.
3. Summarize the concepts of Advanced Trees.
4. Discuss the implementation of various Graph representations and traversals.
5. Outline the basic concepts of Hashing and Collision resolution Techniques.

Course Outcomes

At the end of this Data Structures and Algorithms course, students will be able to:

1. Implement tree traversal algorithms.
2. Apply principles of Binary Search Trees in solving complex problems.
3. Analyze various operations on Advanced Trees.
4. Interpret the importance of Graphs in solving real time applications.
5. Apply the concepts of hashing.

UNIT I

Applications of Stack - Reversing a list, Parenthesis checker, conversion of an infix expression to postfix expression and prefix expression, Recursion, Towers of Hanoi. [Ch 7.7]

Trees: Basic terminology, Types of Trees: General Trees, Binary Trees, Complete Binary Tree, Extended Binary Trees, Representation of Binary Trees in the Memory-Using Arrays and Linked lists (advantages and disadvantages) Traversing a Binary Tree, Constructing a Binary Tree from Traversals. [Ch 9.1,9.2 & 9.4]

UNIT II

Threaded Binary Trees - One-way Threading, Two-way Threading [Ch 10.3], Representation of Algebraic Expressions and its implementation. [Ch 9.2.4]

Binary Search Trees: Operations on BST - searching, insertion, and deletion in BST [Ch10.1 & 10.2]

Heaps: Introduction, Types of Heaps – Min binary heap, Max binary heap. [Ch12.1].

UNIT III

AVL Trees: Operations on AVL Trees- searching, insertion, deletion and rotation [Ch 10.4] . Red-Black Trees-Properties of Red Black Trees and Splay Trees- Operations on splay trees [Ch 10.5,10.6].

B-Trees: Searching, Insertion and deleting in B-trees. [Ch 11.2].

UNIT IV

Graphs: Definition, Graph Terminology, Representation of Graphs: Sequential and linked representation, Graph Traversal Algorithms -Breadth First Search, Depth First Search with algorithms and implementation [Ch 13.1,13.2, 13.5 & 13.6].

Spanning Trees: Definition and its properties, Minimum Spanning Tree, Exploring Minimum Spanning Tree Algorithms: Implementation of Prim's and Kruskal's.

Implementation of Dijkstra Algorithms for finding shortest path in graphs. [Ch13.8]

UNIT V

Hashing: Introduction, Hash Tables, Hash Functions, Collisions: Collision resolution by Open Addressing-Linear probing, Quadratic Probing, Double Hashing, Collision resolution by Chaining. [Ch15:15.1-15.5]

Additional concepts: Rehashing, Extendible Hashing and Implementation of Dictionaries. [Ch 15.5]

Text Books:

1. Reema Thareja, Data Structures using C, Second Edition, Oxford University Press 2014.

Reference Books:

1. Horowitz and Sahani, Fundamentals of Data Structures, Galgotia Publications Pvt Ltd. Delhi India, 2015.
2. Richard F.Gillberg & Behrouz A. Forouzan, Data Structures, A Pseudo code Approach with C, Second Edition, Cengage Learning, India Edition, 2005.

Data Structures and Algorithms Lab

Course Outcomes

At the end of this Data Structures Algorithms Lab course, students will be able to:

1. Develop programs on stacks and its applications
2. Implement operations on Trees
3. Apply the concepts of advanced trees to implement its operations
4. Implement Graph Traversals algorithms
5. Appreciate the role of Hashing and Dictionaries

Week 1

1. Program to check nesting of parentheses using a stack.
2. Program to convert an infix expression into its equivalent postfix notation

Week 2

3. Program to convert an infix expression to a prefix expression.
4. Program to implement towers of Hanoi

Week 3

5. Program to illustrate tree traversals
 - a) In order
 - b) Preorder
 - c) Post order

Week 4

6. Program to implement simple right in-threaded binary trees.

Week 5

7. Program to illustrate insertion, deletion and searching in Binary Search Tree

Week 6

8. Program to implement Heaps: a) Min Heap b) Max Heap

Week 7

9. Program to illustrate Insertion on AVL Trees
10. Program to illustrate deletion and Rotation on AVL Trees

Week 8

11. Program to illustrate Graph traversals
 - a) Breadth First Search
 - b) Depth First Search

Week 9

12. Program to implement
 - a) Prim's algorithm
 - b) Kruskal's algorithm

Week 10

13. Program to Implement Dijkstra algorithm

Week 11

14. Program to implement Hashing and collision resolution techniques

Week 12

15. Program to implement Dictionaries

Week 13 & 14 - Review

Note: The above experiments are for indicative purposes only. However, the concerned faculty member can add a few more experiments in addition to the existing. In such cases the concerned faculty member should get the syllabus approved by the BoS.

Object Oriented Programming through Java

B. Tech II Year I Semester					Dept. of Information Technology			
Code	Category	Hours / Week			Credits	Marks %		
EMA2104	Theoretical & Practical (PC)	L	T	P	C	CIE	SEE	Total
		3	0	4	5	50	50	100

Course Objectives

Objectives of Object-Oriented Programming through Java are to:

1. Impart knowledge of core language features of Java
2. Appraise the concepts of Inheritance
3. Elaborate the use of Exceptions in Java
4. Familiarize concepts of packages and interfaces
5. Emphasize Multi-threading concepts in Java

Course Outcomes

At the end of this Object-Oriented Programming through Java course, students will be able to:

1. Appraise the basic concepts of Java
2. Implement base and derived classes using inheritance
3. Design and develop custom exception classes
4. Implement user-defined packages to support code modularity
5. Apply synchronization to ensure thread safety in shared resources

UNIT I

Introduction to OOPS & Java Basics: History of Java, OOP Principles, Java buzzwords, data types, variables, scope and lifetime of variables, arrays, operators, expressions, control statements, simple java program. [Ch-1, 2, 3, 4 & 5]

(Basic concepts on operators, control statements may be conducted in line with regular programming exercises)

UNIT II

Classes & Methods: Concepts of classes, objects, constructors, methods, access control, this keyword, static keyword, Garbage collection, Overloading methods and constructors, parameter passing.

Inheritance: Inheritance Basics-Member Access and Inheritance, A super class variable can reference a subclass object, types of inheritance, using super keyword, creating a Multilevel Hierarchy, using final with inheritance. [Ch-6, 7, & 8]

UNIT III

Polymorphism: Method Overriding, Dynamic Method Dispatch, Abstract classes, Object class.

Exception Handling: Concepts of exception handling, exception hierarchy, usage of try, catch, throw, throws and finally, built-in exceptions, creating own exception subclasses. [Ch-8 & 10]

UNIT IV

Packages and Interfaces: Defining a package, Access Protection-An Access Example, Importing packages, Interfaces-Implementing interfaces, Nested Interfaces, differences between classes and interfaces, File, Byte Streams, Character Streams. [Ch-9 & 22]

UNIT V

Multi-Threading: Differences between multithreading and multitasking, thread life cycle, creating threads, thread priorities, synchronizing threads, inter thread communication. [Ch-11]

Text Book

1. Herbert Schildt, Java - The Complete Reference, Twelfth Edition, Tata McGraw Hill, 2022.

Reference Books

1. Bruce Eckel, Thinking in Java, Fourth Edition, Prentice Hall.
2. Y. Daniel Liang, Introduction to Java programming, Tenth Edition, Pearson education.

Object Oriented Programming through Java Lab

Course Outcomes

At the end of this Object-Oriented Programming through Java Lab course, students will be able to:

1. Implement basic Java programs using control statements
2. Appraise the core concepts of inheritance
3. Develop applications using packages and interfaces
4. Develop applications to handle runtime errors using exception handling
5. Design and develop multi-threaded applications

List of Experiments

Week 1

1. Write a java program on operator precedence and associativity.
2. Write a java program to illustrate the concept of arrays.

Week 2

3. Write a java program to define a class, define instance methods for setting and retrieving values of instance variables and instantiate its object
4. Demonstrate the use of static keyword and this keyword.

Week 3

5. Write a program to illustrate types of constructors and constructor overloading
6. Write a java program to demonstrate the use of String class and its methods.

Week 4

7. Write a java program to illustrate parameter passing techniques
8. Write a java program to illustrate Recursion and nested class

Week 5

9. Write a java program to demonstrate the use of inheritance.
10. Write a java program to illustrate the use of super and final keywords.

Week 6

11. Write a java program to demonstrate the concept of polymorphism
12. Write a program to illustrate Dynamic method Dispatch

Week 7

13. Write a program to illustrate abstract classes
14. Write a program to illustrate try, catch, throw, throws and finally keywords

Week 8

15. Write a java program to illustrate nested try statement.
16. Write a program to implement the concept of User defined Exceptions

Week 9

17. Write a program to illustrate the use of packages.
18. Write a program to illustrate Interfaces.

Week 10

19. Write a program to illustrate Files.
20. Demonstrate the use of I/O Streams.

Week 11

- 21. Write a program to illustrate Multithreading.
- 22. Write a program to illustrate thread priorities.

Week 12

- 23. Write a program to illustrate Thread Synchronization.
- 24. Write a program to illustrate Inter Thread Communication.

Week 13&14

Review.

Note: The above experiments are for indicative purposes only. However, the concerned faculty member can add a few more experiments in addition to the existing. In such cases the concerned faculty member should get the syllabus approved by the BoS.

Integrated Project - I

B. Tech II Year I Semester					Dept. of Information Technology			
Code	Category	Hours / Week			Credits	Marks %		
EVA2101	Exploratory (PC)	L	T	P	C	CIE	SEE	Total
		0	0	2	1	80	20	100

Evaluation Guidelines for the course Integrated Project-I as mentioned below.

CIE/SEE	Assessment component	When	Marks
CIE	Project Report - Phase I	After the completion of 7 Weeks	40
	Project Report - Phase II	After the completion of 14 Weeks	40
SEE	Viva voce	Semester End	20
Overall	CIE assessments will be taken for 80 marks and SEE for 20 marks		

Operating Systems

B. Tech II Year II Semester					Dept. of Information Technology			
Code	Category	Hours / Week			Credits	Marks %		
EMA2201	Theoretical (PC)	L	T	P	C	CIE	SEE	Total
		3	0	0	3	50	50	100

Course Objectives

Course Objectives of Operating Systems are to:

1. Introduce basic concepts of operating system and operating systems structures.
2. Discuss process management and CPU scheduling algorithms.
3. Demonstrate process synchronization and methods for handling deadlock.
4. Describe memory management Techniques.
5. Explore the File system, system security and protection mechanisms.

Course Outcomes

At the end of this Operating Systems course, students will be able to:

1. Summarize operating systems concepts and structures.
2. Apply process scheduling algorithms.
3. Outline process synchronization and deadlock handling mechanisms.
4. Analyze effectively memory management concepts.
5. Illustrate file system, various protection and security measures.

UNIT I

Operating Systems Overview: What operating systems do, Operating System operations, computing environment [Ch1: 1.1, 1.5, 1.12]

Operating Systems Structures Operating system services, User Operating-System Interface, System calls, Types of System calls, Operating System structure. [Ch2: 2.1 - 2.4, 2.7]

UNIT II

Process Management: Overview, Process scheduling, Operations on processes, Inter process communication. Threads: overview, Multithreading models [Ch3: 3.1 - 3.4 & Ch4: 4.1 - 4.2]

Process Scheduling: Basic concepts, Scheduling criteria, CPU Scheduling [Ch5: 5.1 - 5.3]

UNIT III:

Process Synchronization: Background, The critical section problem, Peterson's solution, Synchronization hardware, Semaphore, Classical problems of synchronization, Monitors. [Ch6: 6.1 - 6.7]

Deadlocks: System model, Deadlock characterization, Methods for handling deadlocks, Deadlock prevention, avoidance, detection, recovery from deadlock. [Ch7: 7.1 - 7.7]

UNIT IV

Memory Management: Background, Swapping, Contiguous memory allocation, Paging, Segmentation [Ch8: 8.1 - 8.4, 8.6]

Virtual Memory Management: Background, Demand paging, Copy-on-write, Page-Replacement, Thrashing [Ch9: 9.1 - 9.4, 9.6].

UNIT V

File System: File concept, Access methods, Directory structure, File-system mounting, Allocation methods, Disk structure, Disk scheduling [Ch10: 10.1-10.4, Ch11:11.4 & Ch12:12.2, 12.4]

Protection: Goals of protection, Principles of protection, Domain of protection, Access matrix, Access Control [Ch17: 17.1-17.4, 17.6]

Security: The Security problem, Program threats, System and Network threats. [Ch-18:18.1 - 18.3]

Text Book:

1. Abraham Silberchatz, Peter B. Galvin, Greg Gagne, Operating System Concepts, 9th Edition, John Wiley, 2016.

Reference Books:

1. D. M. Dharmdhere, Operating Systems – A Concept based Approach, 3rd Edition, TMH, 2017.
2. Andrew S Tanenbaum, Modern Operating Systems, 5th Edition, PHI, 2022.
3. Behrouz A. Forouzan, Richard F. Gilberg, Unix and Shell programming, Cengage Learning, 2009.

Design and Analysis of Algorithms

B. Tech II Year II Semester					Dept. of Information Technology			
Code	Category	Hours / Week			Credits	Marks %		
EMA2202	Theoretical (PC)	L	T	P	C	CIE	SEE	Total
		3	0	0	3	50	50	100

Course Objectives

Course Objectives of Design and Analysis of Algorithms are to:

1. Analyze the asymptotic performance of algorithms
2. Apply the Paradigms and approaches to appreciate the impact of algorithm design in practice
3. Synthesize efficient algorithms in common engineering design situations
4. Analyze complex engineering problems using backtracking
5. Utilize data structures and algorithmic design techniques in solving new problems

Course Outcomes

At the end of this Design and Analysis of Algorithms course, students will be able to:

1. Appreciate divide and conquer paradigm
2. Design greedy paradigm for a given problem
3. Apply dynamic-programming technique for a given scenario
4. Design branch and bound paradigm for real time problems
5. Compare and contrast P and NP problems with examples

UNIT I

Introduction: Algorithm, Pseudo code for expressing algorithms, Performance Analysis-Space complexity, Time complexity, Asymptotic Notation- Big oh notation, Omega notation, Theta notation and Little oh notation, Disjoint Sets- disjoint set operations, union and find operations. [Ch1: 1.1, 1.2, 1.3, Book1] [Ch3: 3.1, Book2]

Divide and conquer: General method, applications-Binary search, Quick sort, Merge sort. [Ch3: 3.1, 3.3, 3.5, 3.6, Book1]

UNIT II

Graphs: breadth first search, depth first search, spanning trees, connected and bi connected components. [Ch6: 6.2-6.4, Book1]

Greedy Method: General method, applications-Job sequencing with deadlines, 0/1 knapsack problem, Minimum cost spanning trees, Single source shortest path problem. [Ch4: 4.1, 4.3, 4.5, 4.6, 4.9, Book1]

UNIT III

Dynamic Programming: General method, Multistage graph, applications-Matrix chain multiplication, Optimal binary search trees, 0/1 knapsack problem, All pairs shortest path problem, Travelling salesperson problem. [Ch5: 5.1-5.3, 5.5, 5.9, Book1]

UNIT IV

Backtracking: General method, applications-n-queen problem, sum of subsets problem, graph coloring, Hamiltonian cycles. [Ch7: 7.1-7.5, Book1]

Branch and Bound: General method, applications -Travelling salesperson problem, 0/1 knapsack problem- LC Branch and Bound solution, FIFO Branch and Bound solution.

[Ch8: 8.1-8.3, Book1]

UNIT V

Lower Bound Theory: Comparison trees, NP-Hard and NP-Complete problems: Basic concepts, non-deterministic algorithms, NP - Hard and NP Complete classes, Clique Decision Problem (CDP), Node cover decision problem. [Ch10: 10.1, Ch11: 11.1, 11.3, Book1]

Text Books:

1. Ellis Horowitz, SatrajSahni and Rajasekharam, Fundamentals of Computer Algorithms, Galgotia publications Pvt. Ltd, Second Edition, 2007.
2. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivert and Clifford Stein, Introduction to Algorithms, Third Edition, PHI Learning Private Limited, Eastern Economy Edition, 2008.

Reference Books:

1. Aho, Ullman and Hopcroft, Design and Analysis of algorithms, Pearson education, Reprint 2002.
2. R.C.T.Lee, S.S.Tseng, R.C.Chang and T.Tsai, Introduction to Design and Analysis of Algorithms: A strategic approach, McGraw Hill, 2005.
3. Allen Weiss, Data structures and Algorithm Analysis in C++, Third Edition, Pearson Education.

Software Engineering

B. Tech II Year II Semester					Dept. of Information Technology			
Code	Category	Hours / Week			Credits	Marks %		
EMA2X03	Theoretical (PC)	L	T	P	C	CIE	SEE	Total
		3	0	0	3	50	50	100

Course Objectives

Course Objectives of Software Engineering course are to:

1. Understand software engineering principles and development models.
2. Elicit and model software requirements.
3. Identify design principles for software components and modelling, including web and mobile app design.
4. Apply user interface design principles and evaluate software quality.
5. Utilize software metrics and cost estimation models to predict project effort.

Course Outcomes

At the end of this Software Engineering course, students will be able to:

1. Compare software development models.
2. Demonstrate software requirements by using Software Requirements Specification.
3. Analyze the computational designs.
4. Design user interfaces using golden rules.
5. Estimate software effort using structured techniques and cost models.

UNIT I

The Nature of Software: The Nature of software, The Changing Nature of software. [Ch1, Book1]

Software engineering: Defining the discipline, the software process, software engineering practice, Software Development myths, how it all Starts. [Ch2, , Book1]

Software Process Structure: A Generic Process Model, Defining Framework Activity, Process Assessment and Improvement [Ch3-3.1, 3.2 and 3.5, Book1]

Process models: Prescriptive Process Models - The waterfall model, Incremental process models, Evolutionary Process model, Evolutionary Process Models, Spiral Model, Concurrent Models [Ch:4 – 4.1, Book1]

UNIT II

Agile Development: What is Agility? Agility and the cost of change, what is an Agile Process? Extreme Programming, Other Agile Process Models, A tool set for the Agile process. [Ch5-5.4,5.5, Book1].

Case Study - Selecting a Process Model Safe Home Part-1 and Part-2 [Ch4-4.1.3 , 4.1.4, Book1]

Requirements Engineering: Functional and non-functional requirements, Requirement Engineering Process Requirements elicitation, Requirement specification, Requirement validation, Requirement Change [Ch4 – Book 2]

UNIT III

Design Concepts: The Design Process, Design concepts, The design model [Ch:12- 12.2,12.3,12.4, Book1]

Modeling Level Design: What is a component? Design class-based components, conducting component level design, Computational level design for Web Apps, Computational level design for Mobile Apps [Ch14 – 14.1 to 14.4, Book1]

UNIT IV

User Interface Design: Golden rules, User Interface Analysis and Design [Ch:15 – 15.1, 15.2, Book1]

Software Testing Strategies: A strategic approach to software testing, Strategic issues, Test strategies for conventional software [Ch22- 22.1 to 22.3, Book1]

Testing Conventional Applications: Software Testing Fundamentals, Internal and External Views of Testing, White-Box Testing, Basis Path Testing, Control Structure Testing, Black-Box Testing [Ch23 – 23.1 to 23.6, Book1].

UNIT V

Estimation for Software Projects: Observations on Estimation, The Project Planning Process, Software Scope and Feasibility, Resources, Software Project Estimation, Decomposition Techniques, Empirical Estimation Models, Estimation for Object Oriented Projects, Specialized estimation Techniques, The Make/Buy Decision. [Ch:33-, Book1]

Text Books:

1. Roger S. Pressman, Software Engineering - A practitioner 's Approach, 8th edition. McGraw Hill International Edition, 2005.
2. Somerville, Software Engineering, 10th Edition, Pearson Education, 2009.

Reference Books:

1. K.K. Agarwal & Yogesh Singh, Software Engineering, New Age International Publishers,3rd edition, 2008
2. James F. Peters, Witold Pedrycz, Software Engineering - an Engineering approach, JohnWiely, 2007.
3. Waman S Jawadekar, Software Engineering Principles and Practice, The McGraw-Hill Companies, 2013.
4. <https://nptel.ac.in/courses/106/105/106105182/>
5. Gene Kim, Jez Humble, Patrics Debois, and Jon Wills, The DevOps Handbook – How to create world-class agility, reliability & Security in technology organizations,2nd Edition

Database Systems

B. Tech II Year II Semester					Dept. of Information Technology			
Code	Category	Hours / Week			Credits	Marks %		
EMA2204	Theoretical & Practical (PC)	L	T	P	C	CIE	SEE	Total
		3	0	2	4	50	50	100

Course Objectives

Course Objectives of Database Systems are to:

1. Discuss Database concepts and apply ER modeling for effective database design.
2. Familiarize the students with relational query languages and foundation of SQL.
3. Enhance the student's ability to Implement intermediate and advanced SQL features.
4. Acquire the knowledge of relational database design and features of transactions.
5. Explore concurrency control methods and understand recovery techniques.

Course Outcomes

At the end of this Database Systems course, students will be able to:

1. Model Entity-Relationship diagrams for enterprise level databases.
2. Formulate Queries using SQL and Relational Formal Query Languages.
3. Develop proficiency in intermediate and advanced SQL.
4. Apply relational database design principles to normalize given database.
5. Implement concurrency control protocols and recovery algorithms to maintain Consistency of database.

UNIT I

Introduction to Database System Concepts: Database-System Applications, Purpose of Database Systems, View of Data, Database Languages, Database Architecture, Database Users and Administrators.

Introduction to the Relation Models and Database Design using ER Model: Structure of Relational Databases, Database Schema, Keys. Overview of the Design Process, The Entity-Relationship Model, Constraints, Removing Redundant Attributes in Entity Sets.

Entity-Relationship diagrams. [Chapter – 1.1 to 1.4, 1.9, 1.12, 2.1 to 2.3, 7.1 to 7.5]

UNIT II

Formal Relational Query Languages: The relational algebra, Tuple Relational Calculus.

Introduction to SQL: Overview of the SQL Query Language, SQL Data Definition, Basic Structure of SQL Queries, Additional Basic Operations, Set Operations, Null Values, Aggregate Functions, Nested Sub queries. Modification of Database. [Chapter- 6.1, 6.2, 3.1 to 3.9]

UNIT III

Intermediate SQL: Join Expressions, Views, Transactions, Integrity Constraints, SQL Data Types and Schemas, Authorization.

Advanced SQL: Functions and Procedures, Triggers, Recursive Queries, Advanced Aggregate Functions. [Chapter – 4.1 to 4.6, 5.1 to 5.5]

UNIT IV

Relational Database Design: Features of Good Relational Designs, Atomic Domains and First Normal Form, Functional Dependency Theory, Closure set of Functional dependencies, second Normal Form, Third Normal Form, Boyce Codd Normal form.

Transactions: Transaction Concept, Simple Transaction Model, Transaction Isolation, ACID Properties, Serializability. [Chapter – 8.1, 8.2, 8.3.2, 8.3.4, 8.4, 8.5, 14.1, 14.2, 14.5, 14.6]

UNIT V

Concurrency Control: Lock-Based Protocols, Deadlock Handling, Multiple Granularity, Timestamp-Based Protocols, Validation-Based Protocols.

Recovery System: Failure Classification, Storage, Recovery and Atomicity, Recovery Algorithm - ARIES. [Chapter – 15.1.1, 15.1.2, 15.2 to 15.5, 16.1 to 16.4, 16.8]

Text Book:

1. Abraham Silberschatz, Henry F. Korth, S. Sudarshan, "Database System Concepts", Sixth Edition, Tata McGraw-Hill 2006.

Reference Books:

1. Raghu Rama Kirshna, Johannes Gchrke, "Database Management System", Third Edition, TATA McGraw Hill, 2003.
2. CJDate, A Kannan, S Swamynathan, "An Introduction to Database Systems", Eight Edition Pearson 2006.

Databases Systems Lab

Course Outcomes

At the end of this Database systems Lab course, students will be able to:

1. Apply SQL commands to perform various operations on a database.
2. Design a database by enforcing integrity constraints.
3. Develop PL/SQL programs using loops and control statements.
4. Design PL/SQL programs using cursors and triggers.
5. Implement PL/SQL procedures and functions

List of Experiments

Week 1

Overview of Database user creation, commands of DDL, DML, DCL and TCL

Week 2

Database Schema for a customer-sale scenario

Customer (Cust_id: integer, cust_name: string)

Item (item_id: integer, item_name: string, price: integer)

Sale (bill_no: integer, bill_date: date, cust_id: integer, item_id: integer, qty_sold: integer)

For the above schema, perform the following—

1. Create the tables with appropriate primary key and foreign key constraints.
2. Insert 10 records in each of the tables.
3. Generate count of items whose item price is greater than 500.
4. List the total Bill details with the quantity sold, price of the item and the final amount.
5. List the details of the customer with their average bill amount.
6. Generate details of item which has maximum item price.
7. Generate details of customer who bought minimum quantity of items

Week 3

Database Schema for a Student Library scenario

Student (Stud_no: integer, Stud_name: string)

Membership (Mem_no: integer, Stud_no: integer)

Book (book_no: integer, book_name: string, author: string)

Iss_rec(iss_no: integer, iss_date: date, Mem_no: integer, book_no: integer)

For the above schema, perform the following—

- a) Create the tables with appropriate primary key and foreign key constraints.
- b) Insert 10 records in each of the tables.
- c) List all student names who have taken a book of 'DBS' or 'Data Structures', by implementing UNION.
- d) List all issues for the current date with student and Book names.
- e) List all student names who have taken a book of 'OS' and 'CD', by implementing INTERSECT.
- f) List the details of students who have a membership but never issued a book, by implementing EXCEPT.
- g) Display a list of books taken by student with stud_no as 5.
- h) List the book details which are issued as of today.

Week 4

Database Schema for a Department-Employee scenario

Departments (dept_no: integer, dept_name: string, dept_location: string);

Employees (emp_id: integer, emp_name: string, emp_salary: integer, dept_no: integer);

For the above schema, perform the following—

- a) Create the tables with appropriate primary key and foreign key constraints.
- b) Insert 10 records in each of the tables.
- c) Develop a query to Grant all privileges on employee table to USER_2.

- d) Develop a query to Grant some privileges on employee table to USER_3.
- e) Develop a query to Revoke all privileges on employee table from USER_2.
- f) Develop a query to Revoke some privileges on employee table from USER_3.
- g) Write a query to implement the save point.
- h) Display the employee name and Department name by implementing a left outer join.
- i) Display the employee name and Department name by implementing a right outer join.
- j) Display the employee id, Department name and Department location by implementing full outer join.

Week 5

Database Schema for a Sailor-Boat scenario

Sailors (sid, sname, rating, age)

Boats (bid, bname, color)

Reserves (sid, bid, day(date))

For the above schema, perform the following—

- a) Create the tables with appropriate primary key and foreign key constraints.
- b) Insert 10 records in each of the tables.
- c) Find the average age of sailors for each rating level by implementing Group By clause.
- d) List all the details of boat reservation in the order of date by using Order By clause.
- e) Find the id and name of sailors who have reserved two different boats on the same day.
- f) Find the id of sailors who have reserved a red boat or a green boat.
- g) Create a view having reserve details of sailors and boats for the last one week.

Week 6

Database Schema for a Customer-Product scenario

Customer (Cust_ID: integer, Cust_Name: string, Email: string)

Product (Product_ID: integer, Product_Name: string, Category: string)

Order (Order_ID: integer, Cust_ID: integer, Product_ID: integer, Order_Date: date, Quantity: integer)

For the above schema, perform the following—

- a) Create the tables with appropriate primary key and foreign key constraints.
- b) Insert 10 records in each of the tables.
- c) Get customer details whose names end with 'n', by implementing LIKE operator.
- d) Display all product names in uppercase, by implementing UPPER function.
- e) Display product name and its length, by implementing LENGTH function.
- f) Concatenate category and product name with a colon, by implementing CONCAT function.
- g) List customer details who ordered a product from the 'Electronics' category.

Week 7

- a) Write a PL/SQL program using loop to count employees with salary above 50000
- b) Write a PL/SQL program using While loop to print name of students who have exactly 5 characters.
- c) Write a PL/SQL program using For loop to update Marks % of all students by adding 20.

Week 8

- d) Write a program to demonstrate the GOTO statement.
- e) Write a program to demonstrate %type and %row type attributes.

Week 9

- a) Write a program to demonstrate predefined exceptions.
- b) Write a program to demonstrate user defined exceptions

Week 10

- a) Create a cursor, which displays names of employees having salary > 50000.
- b) Create a cursor, which displays all employee numbers and names from the EMP table.

Week 11

- a) Create a trigger before/after insert on customer table for each row/statement.
- b) Create a trigger before/after delete on customer table for each row/statement.
- c) Create a trigger before/after update on customer table for each row/statement.

Week 12

- a) Create a function to count number of orders for each customer.
- b) Create a function to find the sum of salaries of all employees working in department number 10.
- c) Create a procedure to display sailor details.
- d) Create a procedure to demonstrate IN, OUT and INOUT parameters.

Week 13 & 14

Review.

Text Book:

Ivan Bayross, SQL, PL/SQL, The programming Language of Oracle, 3rd Revised Edition, BPB Publications, 2008

Note: The above experiments are for indicative purposes only. However, the concerned faculty member can add a few more experiments in addition to the existing. In such cases the concerned faculty member should get the syllabus approved by the BoS.

Advanced Java Programming

B. Tech II Year II Semester					Dept. of Information Technology			
Code	Category	Hours / Week			Credits	Marks %		
EMA2205	Theoretical & Practical (PC)	L	T	P	C	CIE	SEE	Total
		2	0	4	4	50	50	100

Course Objectives

Course Objectives of Advanced Java Programming course are to:

1. Demonstrate the use of various event handling techniques for GUI components
2. Emphasize Graphical User Interface based swing application development
3. Explain different types of JDBC drivers and statement interfaces
4. **Demonstrate** the process of handling HTTP sessions and cookies
5. **Provide** an understanding of JSP

Course Outcomes

At the end of this Advanced Java Programming course, students will be able to:

1. Implement delegation event model in GUI-based applications.
2. Develop Graphical User Interface application through Swings
3. Establish database connections using JDBC
4. Develop client-server application through Java Servlet API
5. Design and develop web applications using JSP

UNIT I

Package java. util: The Collection Interfaces, The Collection classes: LinkedList Class, HashSet Class, TreeSet Class, String Tokenizer, Date, Random, Scanner. [T1, Ch- 20, & 21]

Event Handling: Events, Event sources, Event classes, Event Listeners, Delegation event model, handling mouse and keyboard events, Adapter classes. [T1, Ch-25]

UNIT II

GUI Programming with Swing: Introduction, limitations of AWT, MVC architecture, components, containers. Understanding Layout Managers-Flow Layout, Border Layout, Grid Layout, Card Layout, Gridbaglayout. [T1, Ch 27 & 32]

UNIT III

Exploring Swing: Creating a Swing Application, Painting in Swing, JLabel, JTextField, JButton, JCheckBox, JRadioButton, JScrollPane, JTabbedPane JList, JComboBox [T1, Ch 32 & 33]

UNIT IV

JDBC: JDBC Basics, Different types of JDBC Drivers, Establishing Connection, Executing Statements using Tables, Retrieving Values from Result Sets, Creating a simple application, Using Prepared Statement, Callable Statements. [T2, Ch-26]

Servlets: Overview of Servlets, Architecture of the servlet package, A simple servlet, The lifecycle of a servlet, Running Servlets, Handling GET and Post Requests, using Cookies, Session Tracking. [T2, Ch-34 to 39]

UNIT V

JSP: Introduction to JSP Technology, Understanding the Page Life Cycle, JSP Documents, A Simple JSP Document, JSP Elements-Directives, Actions, Implicit Objects, Scripting Elements. [T3, Ch-6]

Text Books

1. Herbert Schildt, Java - The Complete Reference, 12th edition, Tata McGraw Hill.
2. Campione, Walrath, Huml, The Java Tutorial Continued The Rest of the JDK, 3rd edition, 2008, Pearson education, Asia
3. Kognet Solutions Inc, Java Server programming (J2EE 1.4) Block Book, Dreamtech Press. Platinum Edition

Reference Book

1. Art Gittleman, Ultimate Java Programming Second Edition, dreamtech Press, Scott Jones Publishers.
2. H. M. Dietel, P. J. Dietel Java How to Program Sixth Edition Prentice Hall of India Pvt Ltd, Eastern Economy Edition.

Advanced Java Programming Lab

Course Outcomes

At the end of this Advanced Java Programming Lab course, students will be able to:

1. Implement delegation event model in GUI-based applications
2. Develop GUI-based applications using Swings
3. Connect java applications to database using JDBC
4. Implement HTTP Sessions and cookies using servlets
5. Develop the dynamic web pages using JSP

List of Experiments

Week 1

1. Write a program to illustrate collection classes and interfaces.
2. Write a program to illustrate String Tokenizer, Date, Random and Scanner classes.

Week 2

3. Write a program to illustrate Event Handling (Keyboard and Mouse events).

Week 3

4. Write a java program that works as a simple calculator. Use a grid layout to arrange buttons for the digits and for the +, -, *, % operations. Add a text field to display the result.

Week 4

5. Write a java Program a simple user form which reads the name of a user and mail id in Text fields, select gender with radio buttons, and selects some Known languages using checkboxes, and also enters an address in a text area. After filling details, upon pressing the "submit" button, displays all the information about the user input.

Week 5

6. Write a java Program to create a tabbed pane with two tabs. In the first tab sheet, display some push buttons with names of Branches. In second tab sheet, display checkboxes with names of subjects.

Week 6

7. Write a java program to create a combo box with some name of some places. The user can select any one name from the list and the selected country name is displayed in the frame. (Use JComboBox)
8. Write a java program to select multiple places and displayed in Frame using JList.

Week 7

9. Write a java program that connects to a database using JDBC.
10. Write a JDBC program to create a student table, which includes name, roll no, branch and age or DOB.

Week 8

11. Write a JDBC program to insert and update values into student database.
12. Write a JDBC program to delete values from student database.

Week 9

13. Develop a Servlet application to implement and demonstrate get() and post() methods (Using HttpServlet Class)
14. Write a Servlet program to implement a dynamic HTML using servlet (Username and password should be accepted using HTML and displayed using servlet).

Week 10

15. Write a Servlet program to track HttpSession by accepting user name and password using HTML and display the profile page on successful login.

Week 11

16. Write a program to create cookies and retrieval using servlet.

Week 12

17. Write a JSP program to display HELLO WORLD message along with current date
18. Create a registration form using JSP.

Week 13

19. Create login page using JSP and authenticate the user when he submits the login form using the user's name and password from the database.

Week 14 &15

20. Review

Note: The above experiments are for indicative purposes only. However, the concerned faculty member can add a few more experiments in addition to the existing. In such cases the concerned faculty member should get the syllabus approved by the BoS.

English through Theatre Arts

B. Tech II Year II Semester					Dept. of Information Technology			
Code	Category	Hours / Week			Credits	Marks %		
EAE2221	Practical (HS)	L	T	P	C	CIE	SEE	Total
		1	0	2	2	50	50	100

Course Objectives

Course Objectives of English through Theatre Arts are to:

1. Engage in dialogues, role-plays, and to practice.
2. Study and perform scenes from contemporary plays.
3. Work in teams to devise and rehearse performances.

Course Outcomes

At the end of this English through Theatre course, students will be able to:

1. Communicate Effectively in English.
2. Demonstrate improved fluency and accuracy in spoken and written English.
3. Develop clearer and more expressive speech patterns.
4. Analyze and perform scenes from various English-language plays,
5. Collaborate creatively to perform.

About the Course

Introduction: English through Theatre Arts, a unique and dynamic course designed to enhance English language skills through the powerful medium of theatre. This course integrates the creative and various aspects of theatre arts with language learning, offering an immersive experience that goes beyond traditional classroom methods. Through a variety of theatrical activities, students will develop their speaking, listening, reading, and writing skills while fostering creativity, confidence, and cultural awareness

UNIT I

Focus: Introduction to Drama, Types of Drama, Components of Drama, Monologue, Soliloquy, Dramatic Monologue, Drama Vocabulary.

UNIT II

Art of Articulation: Expressing Emotion through Voice, Stress and Intonation.

UNIT III

Art of Writing: Stages of Script Writing, Tips to Write a Script, Script/ Story Writing.

UNIT IV

Stage Directions: Various sections of a stage, Action and Movement, Role-play of a Short Scene.

UNIT V

Creativity: Writing Short Scenes, Giving Dramatic Reading, performing a Play/Monologue/Soliloquy; from Script to Performance.

Reference Books:

1. Katie Mitchell, The Director's Craft: A Handbook for the Theatre, Routledge, 2018.
2. Christopher B. Balme, The Cambridge Introduction to Theatre Studies, CUP, 2018.
3. Edwin, Wilson, "The Theatre Experience" McGraw-Hill Education, 2018.
4. Edwin Wilson and Alvin Goldfarb. Theatre: The Lively Art, McGraw-Hill, 2018.
5. Peter, Brook, The Empty Space, Touchstone Publications, 2020.

Integrated Project - II

B. Tech II Year II Semester					Dept. of Information Technology			
Code	Category	Hours / Week			Credits	Marks %		
EVA2201	Exploratory (PC)	L	T	P	C	CIE	SEE	Total
		0	0	2	1	80	20	100

Evaluation Guidelines for the course Integrated Project-II as mentioned below.

CIE/SEE	Assessment component	When	Marks
CIE	Project Report - Phase I	After the completion of 7 Weeks	40
	Project Report - Phase II	After the completion of 14 Weeks	40
SEE	Viva voce	Semester End	20
Overall	CIE assessments will be taken for 80 marks and SEE for 20 marks		